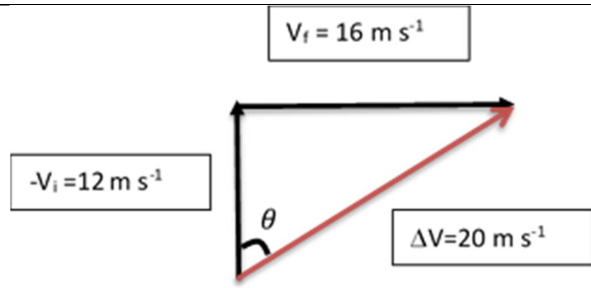
Mark Scheme

1 (a)		C0
	Change in Velocity = Final V – Initial V Magnitude = $\sqrt{(16^2 + 12^2)}$ = 20	A1
	Direction, θ = $\tan^{-1}$ (16/12) = 53.13˚ = 53.1°	A1
(b)	This is because as different pressures are applied in closing the gap of the vernier calipers, the readings will be **inconsistent and fluctuate around the average reading.**	M1
	Random error.	A1
	OR	
	This is because the pressure applied can be consistently more than what is necessary hence resulting in a reading **consistently less than the actual reading.**	M1
	Systematic error.	A1
(c)(i)	Units of e = $\dfrac{units\ of\ P}{units\ of\ (\sigma A T^4)}$ $= \dfrac{W}{(Wm^{-2}K^{-4})(m^2)(K^4)}$	C1
	= 1 (OR No units)	A1
(ii)	e = $\dfrac{P}{\sigma A T^4} = \dfrac{P}{\sigma(\pi \frac{d^2}{4})T^4}$	C1
	$= \dfrac{3}{(5.67x10^{-8})(\pi \frac{0.05^2}{4})500^4}$ = 0.4312 $\dfrac{\Delta e}{e} = \dfrac{\Delta P}{P} + 2\dfrac{\Delta d}{d} + 4\dfrac{\Delta T}{T} = \dfrac{0.2}{3} + 2(\dfrac{0.1}{5}) + \dfrac{4}{500}$	M1
	Δe = (0.115)(0.431) = 0.05 e ± Δe = (0.4312 ± 0.05) (mark is for calculating e and Δe correctly)	C1
	Therefore, e ± Δe = (0.43 ± 0.05) (Mark for correct decimal places)	A1

2	**Concepts tested:** Work, Energy, Power	